

Cambridge International AS & A Level

CANDIDATE
NAME

CENTRE
NUMBER

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FURTHER MATHEMATICS

9231/33

Paper 3 Further Mechanics

May/June 2023

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 m s^{-2} .

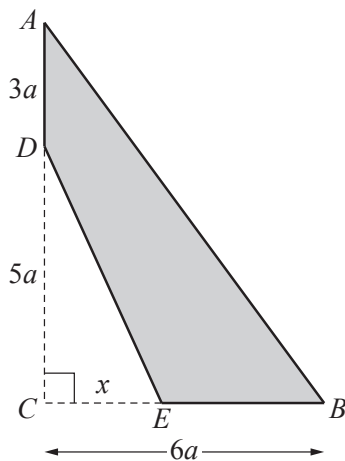
INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

[illegible]

[4]



A uniform lamina is in the form of a triangle ABC , with $AC = 8a$, $BC = 6a$ and angle $ACB = 90^\circ$. The point D on AC is such that $AD = 3a$. The point E on CB is such that $CE = x$ (see diagram). The triangle CDE is removed from the lamina.

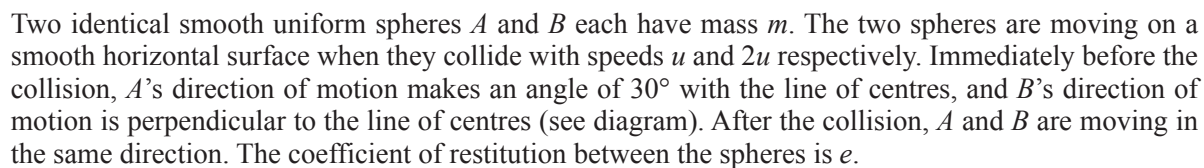
- (a) Find, in terms of a and x , the distance of the centre of mass of the remaining object $ADEB$ from AC . [4]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.

The object $ADEB$ is on the point of toppling about the point E when the object is in the vertical plane with its edge EB on a smooth horizontal surface.

- (b) Find x in terms of a . [3]

[illegible]



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- This image shows a full page of a handwriting practice worksheet. It consists of multiple rows of horizontal dashed lines spaced evenly down the page, providing a guide for letter height and placement. The background is plain white, and there are no margins or additional markings.

- (b)** Find the loss in the total kinetic energy of the spheres as a result of the collision. [3]

[illegible]

- [illegible]

[Turn over

- 6 A particle of mass m kg falls vertically under gravity, from rest. At time t s, P has fallen x m and has velocity v ms⁻¹. The only forces acting on P are its weight and a resistance of magnitude $kmgv$ N, where k is a constant.

(a) Find an expression for v in terms of t , g and k . [5]

This image shows a full page of a handwriting practice worksheet. It consists of multiple rows of horizontal dashed lines spaced evenly down the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.

- (b)** Given that $k = 0.05$, find, in metres, how far P has fallen when its speed is 12 m s^{-1} . [5]

This image shows a full page of a document template designed for handwriting practice or as a guide for letter height. It consists of approximately 20 evenly spaced horizontal dotted lines running across the entire width of the page. The background is plain white, and there are no margins, text, or other markings present.

- [6]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

- (b)** Find the direction in which the ball is moving immediately before the collision. [3]

[illegible]

[illegible]

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